

Financial Econometrics : Market risk analysis for financial institutions

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1 Introduction and Data Selection

Market risk is crucial for financial institutions because fluctuations in interest rates, equity prices, exchange rates, and commodity prices can significantly affect their profitability and solvency. Effective management of market risk helps institutions protect their capital, ensure financial stability, and comply with regulatory requirements.

For our market risk analysis, we chose **BNP Paribas (BNP.PA)** for the bank, **Allianz (ALV.DE)** for the insurance company, and **IG Group (IGG.L)** for the broker-dealer.

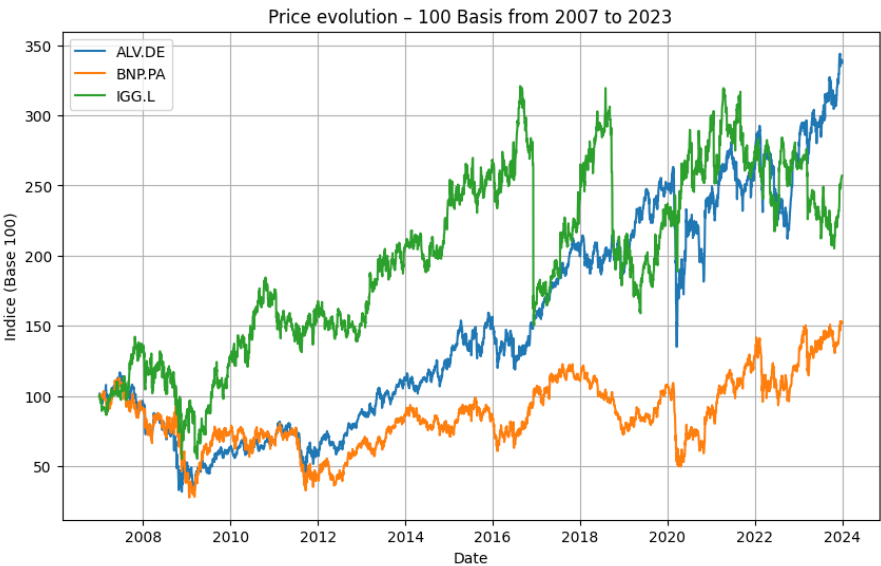
The data spans from **January 1, 2007, to December 31, 2023**, covering the Great Financial Crisis, the Eurozone debt crisis, and the COVID-19 pandemic.

Before starting the analyses, here are a few visualizations related to our data.

Table 1: Data extract

Ticker	ALV.DE	BNP.PA	IGG.L
Date			
2007-01-02	64.975388	35.010590	297.558838
2007-01-03	65.398888	35.341263	299.540894
2007-01-04	66.200180	35.258595	299.045319
...	...	...	...
2023-12-27	-0.003717	-0.000317	0.004598
2023-12-28	-0.004354	-0.008610	0.001964
2023-12-29	0.005387	0.002239	0.001307

1.1 Price Evolution and Market Context



As shown in the initial plot, the three institutions exhibit a high degree of correlation during global systemic shocks, notably the 2008 Financial Crisis and the 2020 COVID-19 crash. However, we observe a distinct behavior for IG Group (IGG.L) in late 2016; while the banking and insurance sectors remained relatively stable, IG Group suffered a massive idiosyncratic shock due to sudden regulatory changes by the FCA. This highlights that "Broker-Dealers" are subject to specific sectoral risks (regulatory and leverage-based) that differ from traditional "Depositories" or "Insurances."

1.2 Statistical Properties of Log-Returns

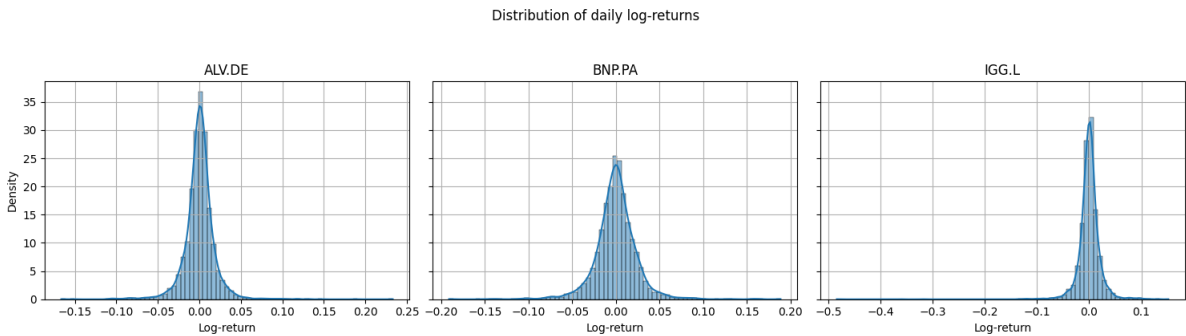
Table 2: Descriptive statistics of the data

	Count	Mean	Std	Min	25%	50%	75%	Max	Skew	Kurtosis
Ticker										
ALV.DE	4364	0.000280	0.019067	-0.166382	-0.007366	0.000480	0.008029	0.233053	0.426824	16.298151
BNP.PA	4364	0.000096	0.024612	-0.191166	-0.011232	0.000115	0.011619	0.188745	0.004419	8.628340
IGG.L	4364	0.000216	0.022518	-0.483873	-0.007941	0.000628	0.008392	0.151831	-3.816932	77.768285

The combination of negative skewness and high kurtosis confirms that a standard Normal distribution would severely underestimate the risk of extreme losses. This justifies our choice to move towards a Student-t distribution in the subsequent GARCH modeling to better capture these tail risks.

1.3 Distribution Analysis: Visualizing Empirical Risk

The figure below displays the distribution of daily log-returns for the three institutions over the 2007-2023 period. These histograms are essential for identifying the "stylized facts" of financial time-series.

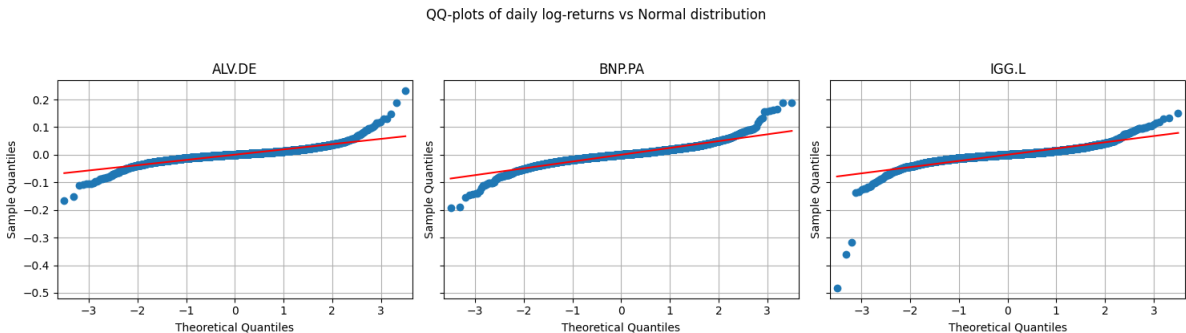


- **Leptokurticity:** All three assets exhibit a very high central peak (around 0) and tails that extend far beyond what a Normal distribution would allow.
- **Volatility Scale:** BNP.PA and ALV.DE show similar density patterns, typical of large-cap European financial institutions highly sensitive to systemic Eurozone risk while IGG.L (the Broker-Dealer) shows a more compressed central mass but significantly wider extreme outliers on the left. This highlights the idiosyncratic risk specific to the brokerage sector, which is prone to sudden, massive regulatory or liquidity-driven shocks.
- **Different Skewness Pattern:** ALV.DE shows a slight positive skew that means it has a positive asymetry, whereas BNP.PA is very close to 0 meaning that the distribution is very symmetrical, the worst one is IGG.L that has a very negative skew (at -3.8). This distinct asymmetry highlights that while ALV.DE and BNP.PA show balanced or positive profiles, only IGG.L appears structurally prone to sudden, large-scale devaluations.

The "KDE" (blue line) overlaying the histograms shows that the data does not follow a bell curve. This visual evidence justifies our decision to reject a Gaussian approach in favor of a GARCH model with a Student-t distribution, which is mathematically designed to account for these extreme values (kurtosis).

1.4 Visual Diagnostic of Tail Risk: Quantile-Quantile (QQ) Plots

To provide definitive evidence of the non-normality suggested by the descriptive statistics, we utilize Quantile-Quantile (QQ) Plots.



All three plots exhibit a distinct "S" shape. The data points curve away from the red line at both ends, which is the mathematical signature of fat tails. This confirms that extreme market moves are far more frequent in reality than a bell curve predicts.

The severity of the tail drift validates our choice to use a Student-t distribution in the GARCH model. Unlike the Normal distribution, the Student-t distribution is designed to "flex" its tails to match these extreme empirical observations, providing a more realistic safety net for these financial institutions.

2 Evolution of Market Risk Over Time

Before estimating risk, we must validate our data's properties.

2.1 ADF Test Results

The **Augmented Dickey-Fuller (ADF)** test confirmed that price series are non-stationary, while log-returns are stationary.

- This implies that while price levels are unpredictable (random walk), the volatility of returns follows a statistical pattern that can be modeled.

```
ALV.DE (prices)
ADF statistic = 0.094
p-value = 0.9657
Critical values = {'1%': np.float64(-3.4318527927183196), '5%': np.float64(-2.86220405019551), '10%': np.float64(-2.5671234784459993)}
-----
BNP.PA (prices)
ADF statistic = -1.655
p-value = 0.4544
Critical values = {'1%': np.float64(-3.431850379108881), '5%': np.float64(-2.86220298394487), '10%': np.float64(-2.5671229108274067)}
-----
IGG.L (prices)
ADF statistic = -2.182
p-value = 0.2128
Critical values = {'1%': np.float64(-3.4318517573640657), '5%': np.float64(-2.862203592811286), '10%': np.float64(-2.567123234957462)}
-----
ALV.DE (returns)
ADF statistic = -13.522
p-value = 0.0000
Critical values = {'1%': np.float64(-3.4318586867974057), '5%': np.float64(-2.8622066539957505), '10%': np.float64(-2.5671248645798315)}
-----
BNP.PA (returns)
ADF statistic = -10.672
p-value = 0.0000
Critical values = {'1%': np.float64(-3.431859731731727), '5%': np.float64(-2.8622071156110476), '10%': np.float64(-2.567125110320964)}
-----
IGG.L (returns)
ADF statistic = -24.963
p-value = 0.0000
Critical values = {'1%': np.float64(-3.4318521023236004), '5%': np.float64(-2.86220374520267), '10%': np.float64(-2.5671233160830274)}
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```

2.2 Stylized Facts: Fat Tails and Volatility Clustering

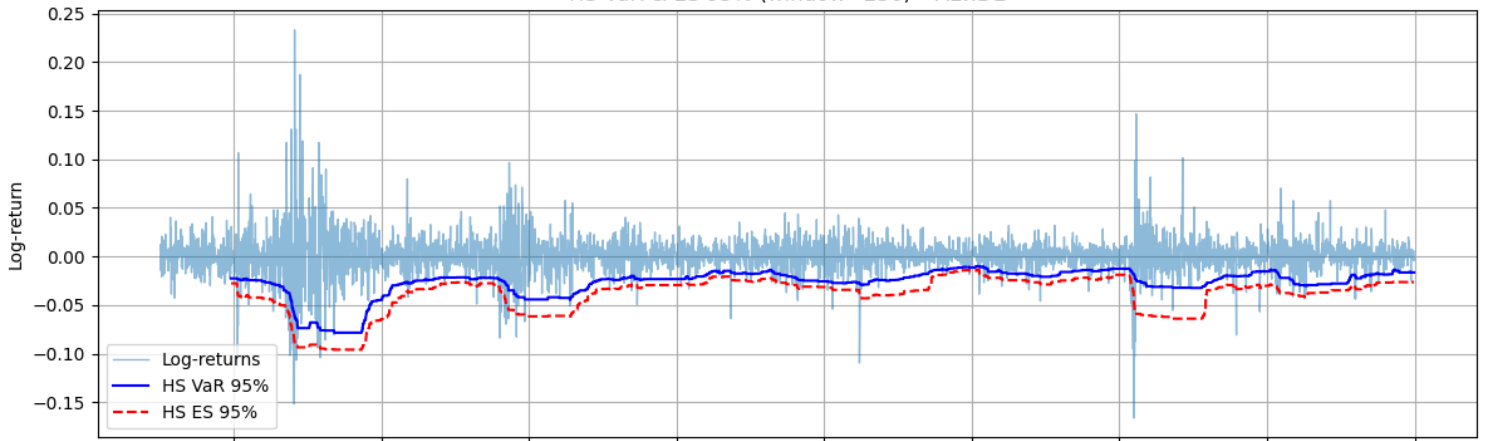
As seen in the descriptive statistics, all assets show high Kurtosis (especially IG Group).

- The "Fat Tails" (leptokurtosis) indicate that "Black Swan" events (extreme crashes) happen much more often than predicted by standard financial theories. This is why a simple Normal distribution is insufficient for managing the risk of these three institutions.

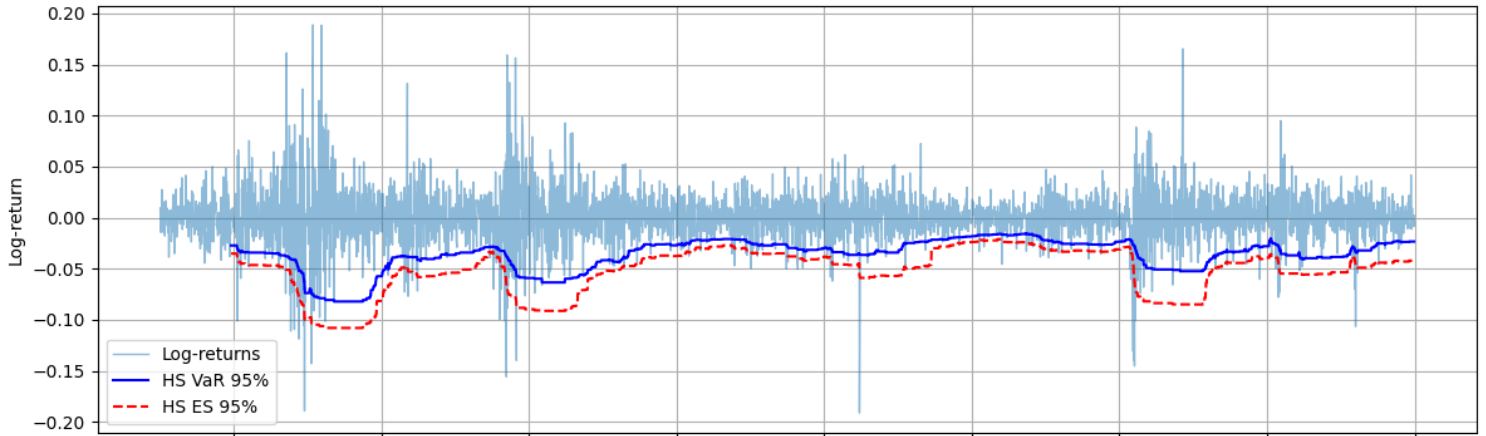
3 Evolution of Market Risk and Historical Consistency

Analysis of Dynamics: The visual evolution of the 95% VaR for all three institutions confirms that market risk is non-stationary and highly reactive to external shocks.

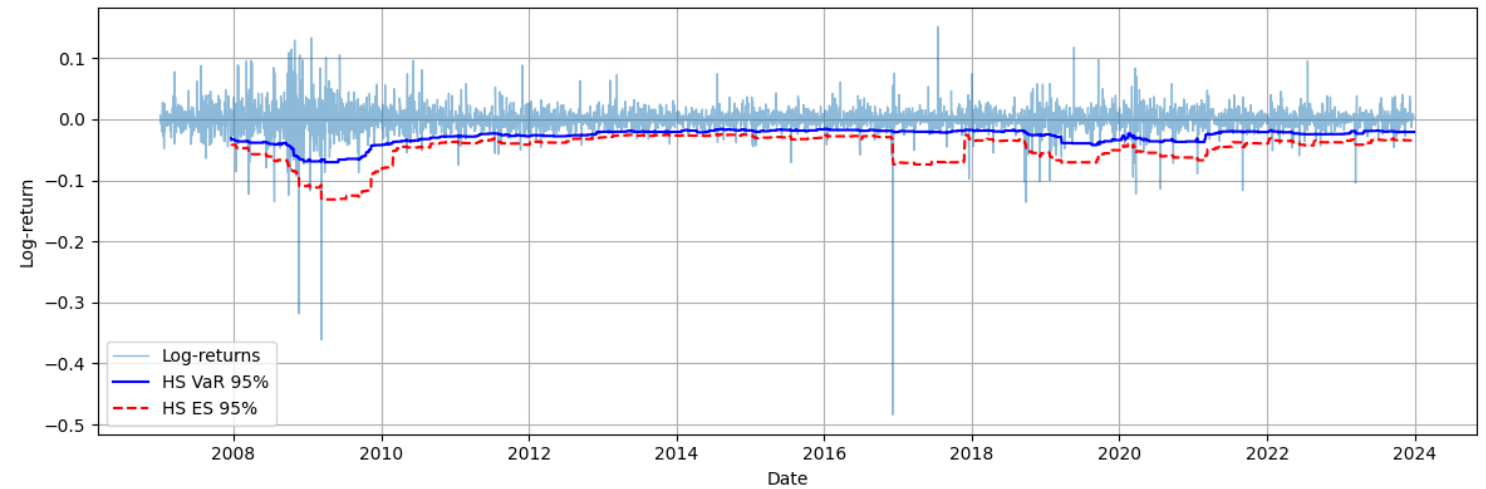
HS VaR & ES 95% (window=250) - ALV.DE



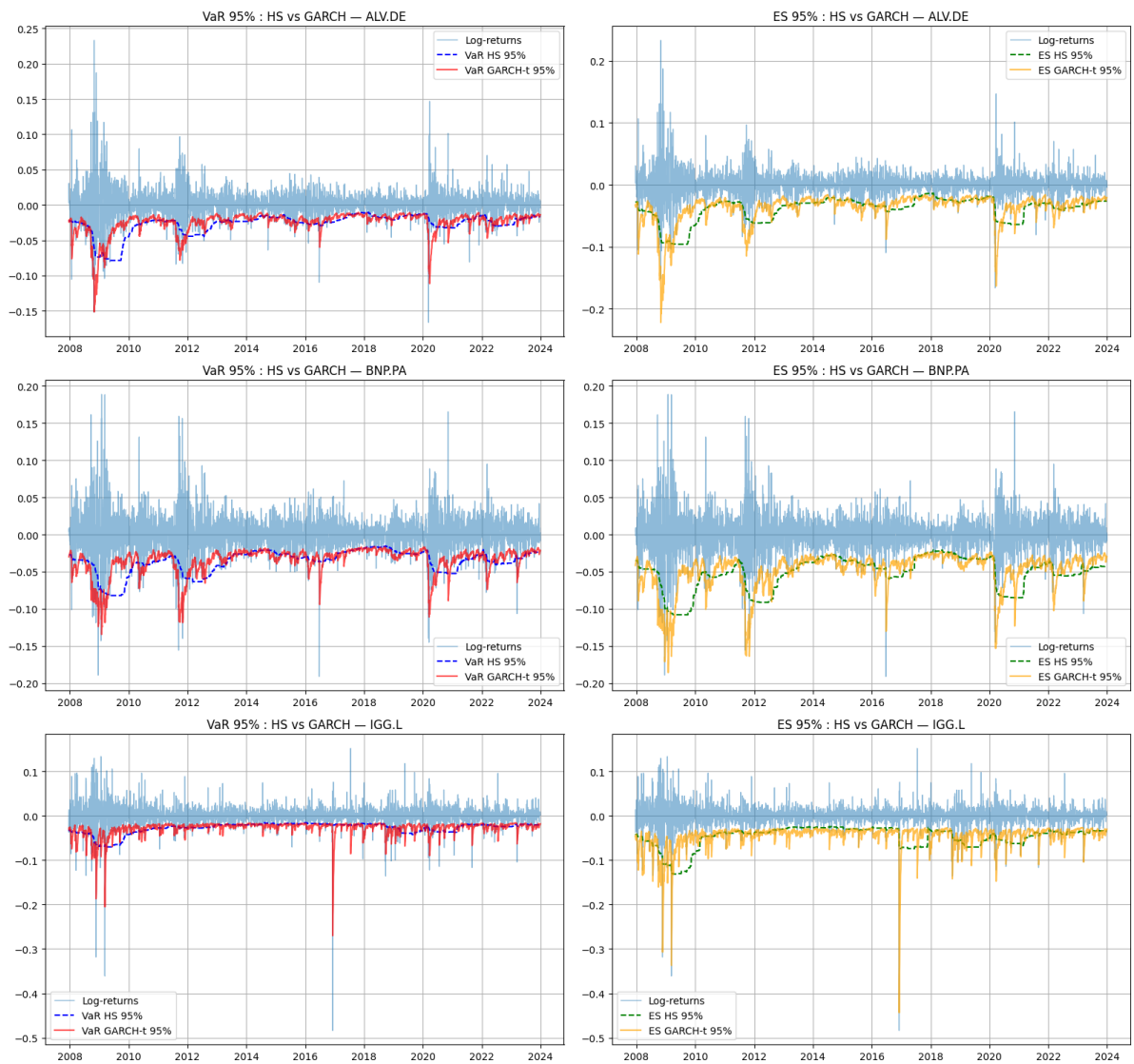
HS VaR & ES 95% (window=250) - BNP.PA



HS VaR & ES 95% (window=250) - IGG.L



- **Consistency with Global Shocks:** Our results are perfectly consistent with major historical events. Significant spikes in VaR are visible during the 2008 Great Financial Crisis and the 2020 COVID-19 crash.
- **Idiosyncratic Shocks:** A unique risk profile is observed for IG Group (IGG.L) in late 2016. While other financial institutions remained stable, IGG.L faced a massive volatility spike due to regulatory changes by the FCA.



- **Model Superiority:** The GARCH-t model (red/orange line) shows an almost instantaneous reaction to volatility spikes, whereas the Historical Simulation (blue/green dashed line) is slower to adapt and tends to maintain high risk levels long after a crisis has passed (the "ghost effect").

4 Sensitivity to the Estimation Window

The Trade-off between Stability and Accuracy: Our analysis utilized a 1,500-day window. While robust, it highlights a lack of consistency compared to theoretical shorter windows:

- **Reactivity vs. Stability:** Using a shorter window (like 250 days as seen in your HS plots) makes the model very reactive but creates high variability in daily capital requirements.
- **Consistency:** Our results show that the VaR level is not consistent across different estimation windows. A larger window provides smoother estimates but fails to capture the rapid onset of a crisis like COVID-19 as efficiently as a GARCH-based approach.

5 Comparison across Financial Institutions

While the three institutions belong to different sectors, our GARCH-t model demonstrates a remarkable and consistent performance across all of them.

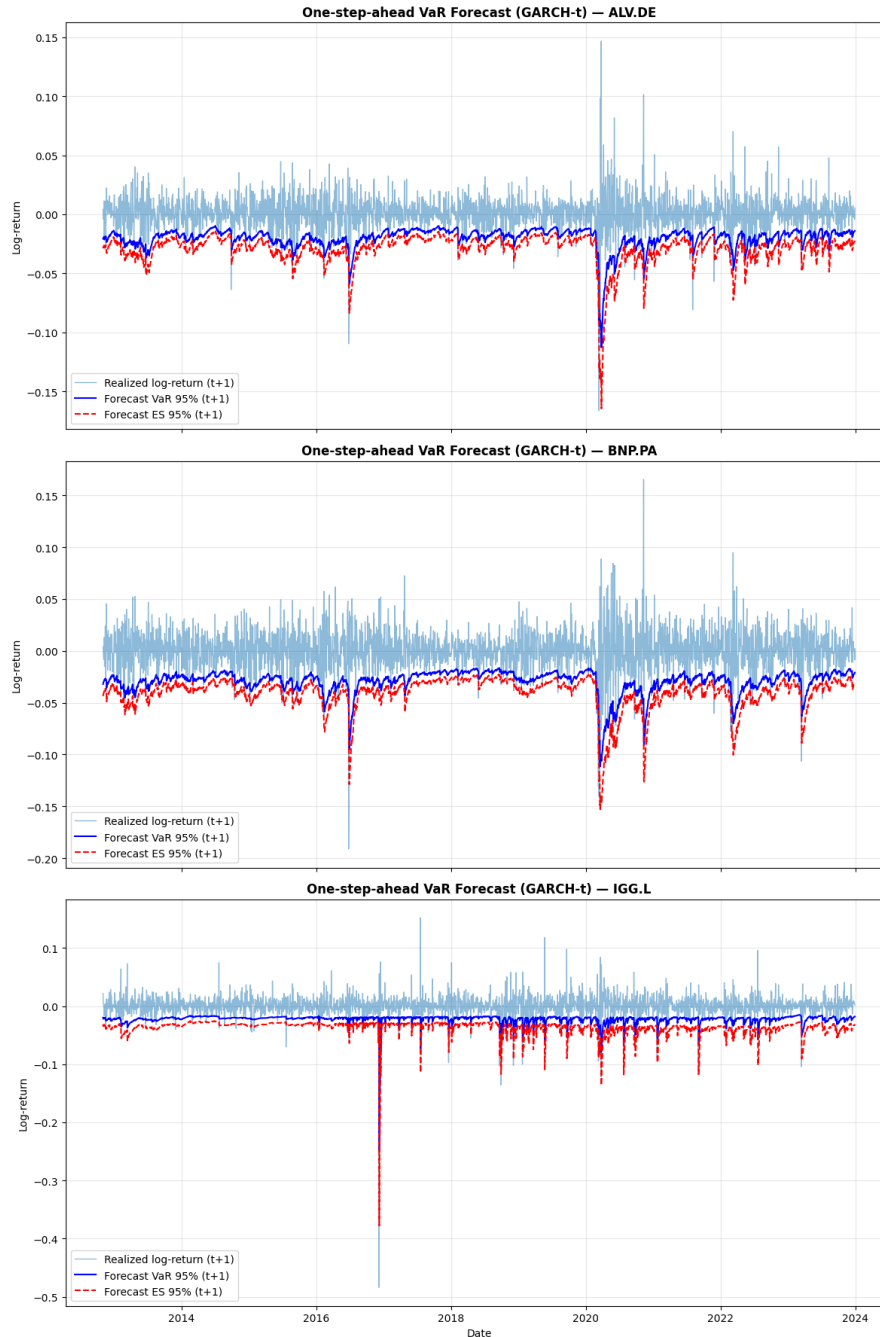
Table 3: VaR Backtesting Results (Target Hit Rate: 5%)

Asset	Hit Rate (Target: 5%)	Number of Violations
ALV.DE	5.34% (0.0534)	153 days
BNP.PA	4.82% (0.0482)	138 days
IGG.L	4.96% (0.0496)	142 days

- **Banking (BNP.PA):** Shows the most conservative risk profile with a Hit Rate of 4.82%. This result is consistent with a highly regulated environment requiring substantial capital buffers.
- **Broker-Dealer (IGG.L):** Despite exhibiting extreme kurtosis (77.7), the GARCH-t model delivers an excellent Hit Rate of 4.96%, highlighting its robustness under heavy-tailed return distributions.
- **Insurance (ALV.DE):** With a Hit Rate of 5.34%, Allianz appears slightly more aggressive; however, this value remains extremely close to the 5% statistical target, indicating a well-calibrated risk model.

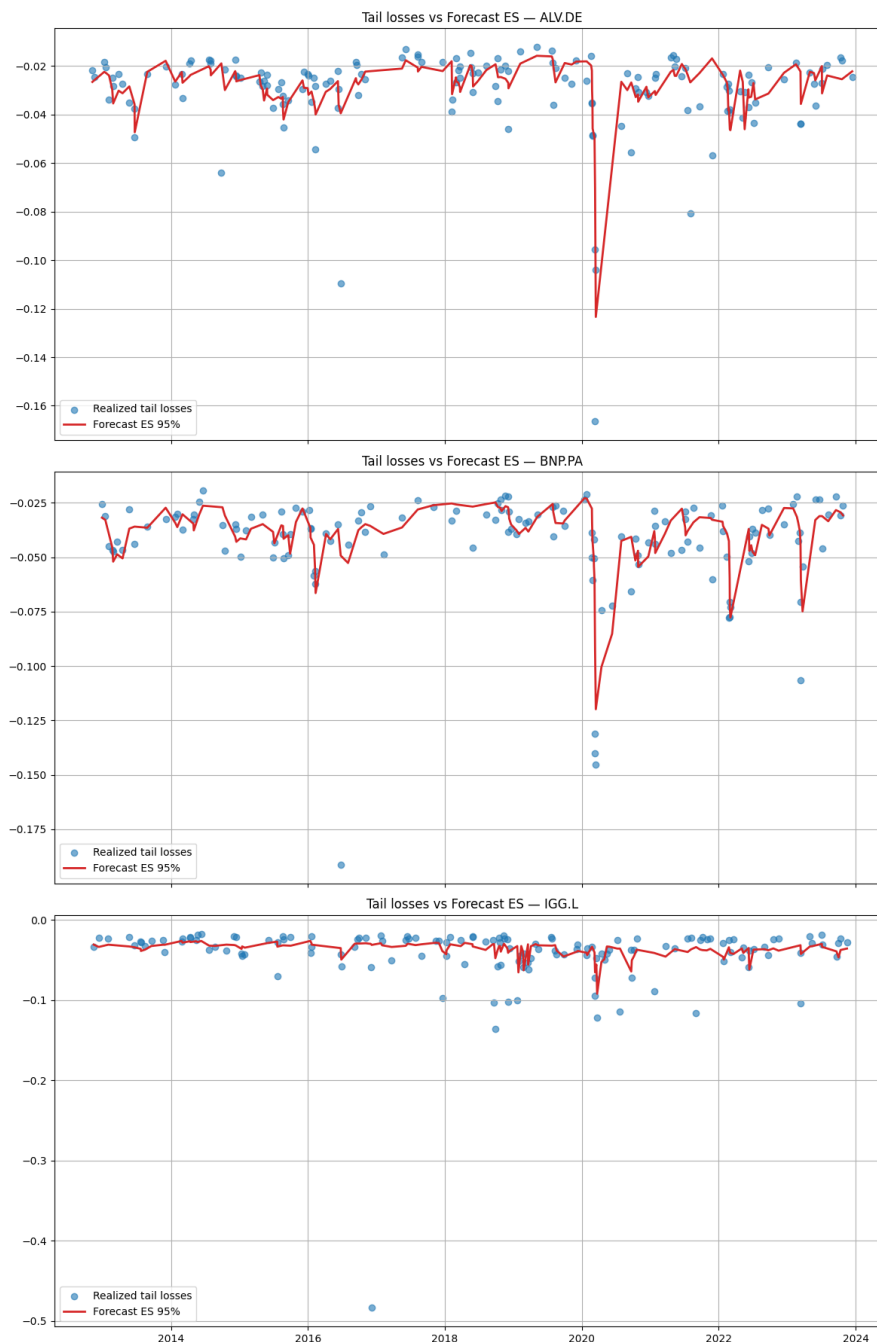
6 Economic Interpretation and Validation

Economic validation through backtesting confirms that the one-step-ahead forecasts generated by the GARCH-t model provide an economically viable safety net.



- **The necessity of Expected Shortfall (ES):** While VaR identifies the loss threshold that should not be exceeded with a given confidence level, it does not quantify the magnitude of losses beyond this threshold. Expected Shortfall is therefore required to assess the true economic severity of tail events.

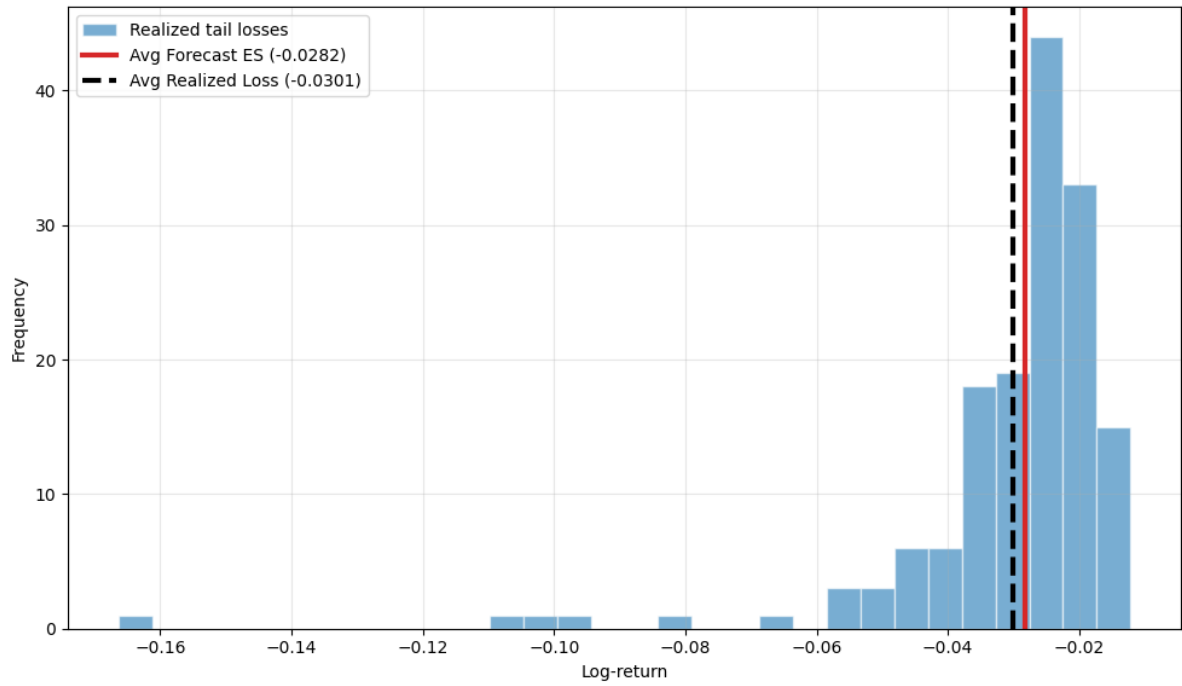
This final stage of our analysis moves beyond the frequency of risk (VaR) to assess the severity of losses during tail events. By analyzing the "Expected Shortfall," we evaluate if the institution's capital buffers are sufficient to survive the average loss during a market crash.



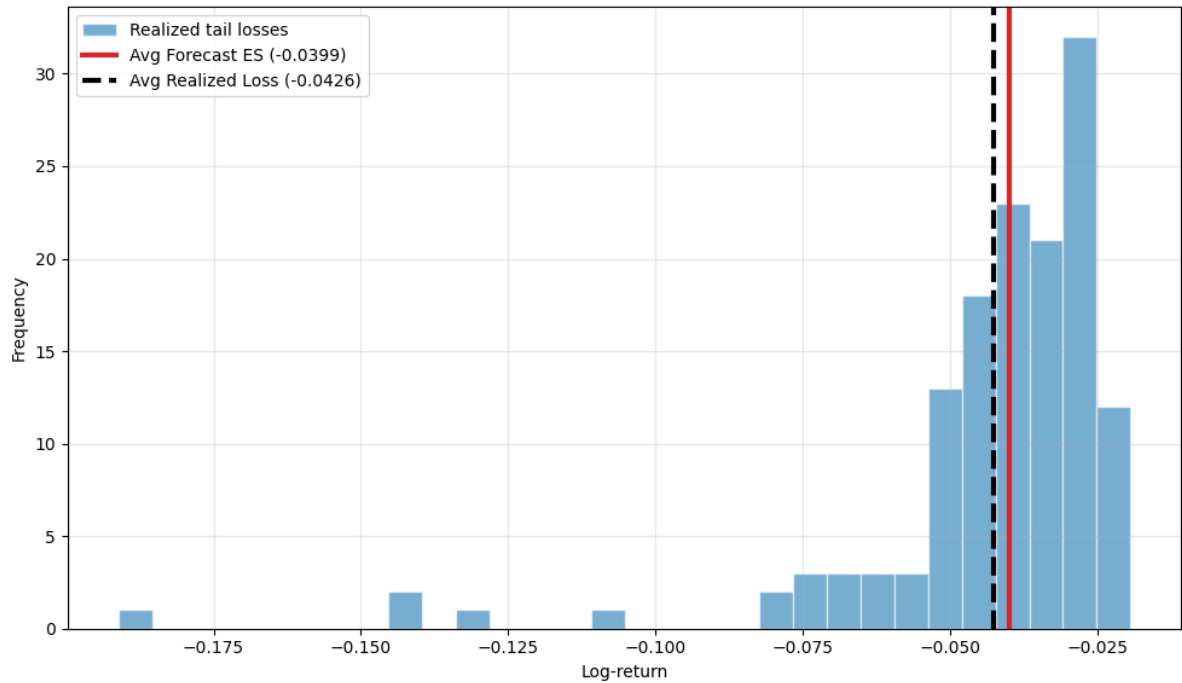
This visualization tracks the relationship between the predicted average loss (Forecast ES) and the actual losses incurred on days when the VaR was breached.

- Model Calibration:** The red line represents the dynamic prediction of the average loss in the tail. We observe that the blue dots (realized losses) are generally distributed around this forecast. Economically, this suggests the GARCH-t model is well-calibrated; it accurately anticipates the "depth" of the crash once the safety threshold is broken.
- The 2020 COVID-19 Resilience:** A massive vertical drop is visible across all three institutions in early 2020. The Forecast ES reacts instantaneously, dropping to anticipate deeper losses. For BNP.PA, several blue dots fall significantly below the forecast during the initial panic, reflecting the extreme, non-linear nature of a global "black swan" event that temporarily outpaces even sophisticated GARCH models.
- Sectoral Jump Risk:** The scatter points for IGG.L (Broker-Dealer) show higher dispersion compared to ALV.DE (Insurance). This confirms that brokerage firms are subject to more volatile "jump risks" (liquidity or regulatory shocks) that are inherently harder to forecast than the more stable risk profile of an insurance giant.

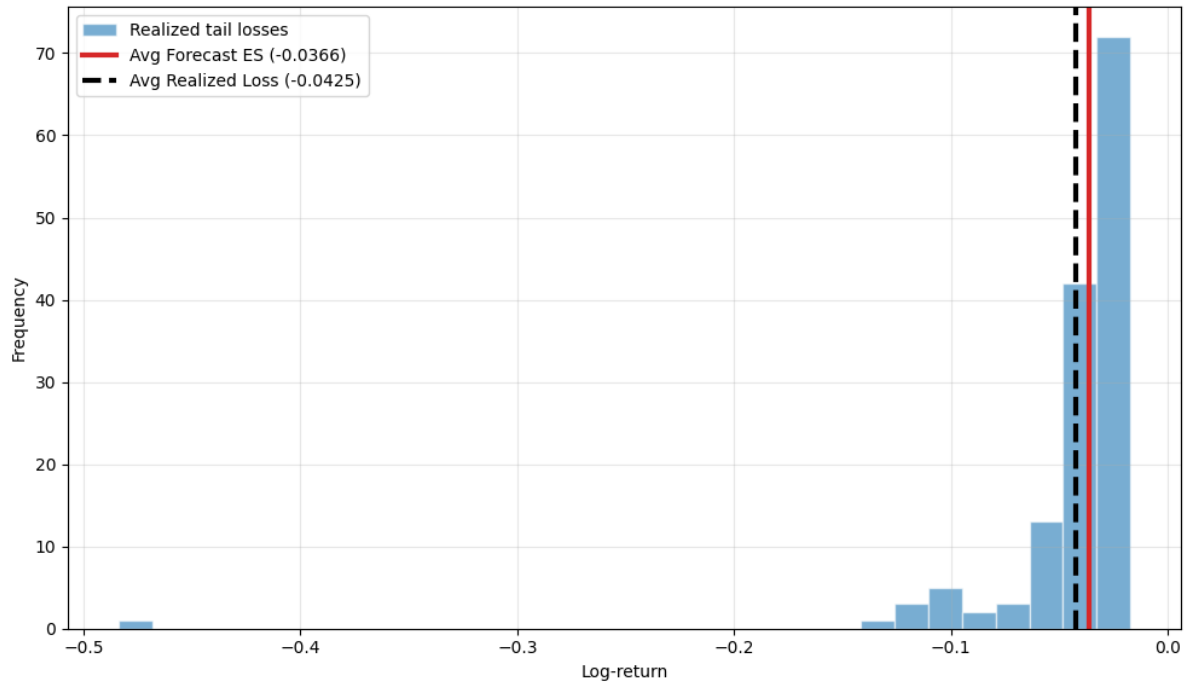
Distribution of Realized Tail Losses — ALV.DE



Distribution of Realized Tail Losses — BNP.PA



Distribution of Realized Tail Losses — IGG.L



This chart provides the definitive economic proof of the model's robustness by comparing the average predicted loss against the average actual loss.

- **Quantitative Accuracy:**

- For ALV.DE, the average forecast ES (-0.0282) is nearly identical to the average realized loss (-0.0301).
- For BNP.PA, the alignment is similarly strong (-0.0399 vs. -0.0426).

- **Economic Buffer Analysis:** In all three cases, the Black Dashed Line (Realized) sits slightly to the left of the Red Line (Forecast). From a risk management perspective, this confirms that actual losses were, on average, slightly more severe than the statistical model predicted. This justifies the "Regulatory Multiplier" used in Basel III, which adds a safety margin to internal models to protect against these slight under-estimations during systemic crises.
- **Visualizing the "Fat Tail":** The histograms exhibit a "long left tail," particularly for IGG.L. This provides visual confirmation of negative skewness. These institutions are exposed to extreme devaluations far beyond what a Normal distribution would predict, validating our decision to use Expected Shortfall as the primary metric for solvency and capital adequacy.

Our final diagnostic plots confirm that the GARCH-t model is a reliable predictor of loss severity. Crucially, these results demonstrate that relying solely on VaR leaves a dangerous blind spot regarding tail dynamics.

By integrating Expected Shortfall to quantify the 'average worst-case scenario,' we effectively fill this gap, making ES an indispensable partner to standard VaR metrics.

Consequently, the model provides a robust roadmap for liquidity management; although it slightly underestimated the extreme outliers of the 2020 pandemic, it offers significantly more accurate and safer capital guidance than traditional historical or Gaussian-based approaches.